

# Muqarrab Ahmed

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## SUMMARY

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I am a Computer Science undergraduate at **FAST NUCES Islamabad**, with a strong foundation in problem-solving and a keen interest in **Artificial Intelligence and Machine Learning**. I have basic hands-on experience with Python and PyTorch through academic and research projects, and I am eager to apply my learning to real-world AI/ML challenges while continuously building my skills.

## PORTFOLIO

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- **AI-Powered Real Estate SaaS Platform (Real Assist)**

[View website](#)

Developed an complete AI powered real estate SaaS platform for ABS Developers, integrating intelligent property discovery and automation features. Implemented **Retrieval-Augmented Generation (RAG)** to enable semantic property search and document-based query answering. Added **text-to-speech (TTS)** capabilities for accessibility and AI-driven user interaction.

*Tech Stack: React.js, Tailwind CSS, Node.js, VectorDB, RAG, AI APIs, Render*

- **NutriLens — AI-Based Food Image Classification App**

[View on Github](#)

Designed and developed an AI powered mobile application for fitness and Gym professionals to help them in their diet, it classifies food images to assist users in nutrition tracking and diet planning. Trained an **EfficientNet-B0** convolutional neural network (5M parameters) on **Food-101 and custom food image datasets**, achieving **87% classification accuracy**.

*Tech Stack: Flutter, TensorFlow, EfficientNet-B0, Python, Computer Vision, Food-101 Dataset*

- **Autonomous Racing Agent Using Deep Neural Networks (Research Project)**

Collected **50,000+ rows of telemetry data** (steering, throttle, braking, tire grip, etc.) from manual gameplay across multiple laps and cars. Trained a **PyTorch neural network** to predict control actions, enabling the vehicle to run **fully autonomously**.

*Tech Stack: PyTorch, Neural Networks, Python, Supervised Learning, CSV Datasets, Game AI*

## WORK EXPERIENCE

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### AI/ML Engineer Apprentice at Finova

June 2025 – Present

- Assisted in training and fine-tuning **machine learning models** for classification and NLP tasks.
- Integrated **Text-to-Speech (TTS)** and **Speech-to-Text (STT)** models for voice-enabled applications.
- Implemented **Gemini File Search API** to build document-aware AI assistants.
- Applied **Hugging Face pre-trained models** for text understanding, summarization, and semantic search, validated using **Kaggle** datasets.

### Freelance AI/ML Projects (Entry-Level)

Aug 2024 – Present

- Built small-scale **AI-driven features** such as automated classification and intelligent search.
- Developed proof-of-concept solutions using **pre-trained models** for generation and classification tasks.
- Prepared **datasets** through cleaning, labeling, and validation for ML workflows.
- Conducted data preprocessing and exploratory analysis using **Python** on **Google Colab**.

# EDUCATION

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|----------------|---------------------------------------------------------------|
| 2022 – Present | Bachelor’s in Computer Science at <b>FAST Nuces Islamabad</b> |
| 2020 – 2022    | Intermediate Federal Board at <b>Hamza APS and College</b>    |
| 2018 – 2020    | Matric Federal Board at <b>SLS Montessori and High School</b> |

# SKILLS

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|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Machine Learning &amp; AI:</b> | Image Classification, EfficientNet, Retrieval-Augmented Generation (RAG), Model Training, Model Inference, AI Automation, pre-Trained model Fine Tuning, TTS and STT |
| <b>Programming Languages:</b>     | Python, JavaScript, TypeScript, C++, C                                                                                                                               |
| <b>Frontend Development:</b>      | React.js, Tailwind CSS, Flutter, Responsive UI Design                                                                                                                |
| <b>Backend &amp; Databases:</b>   | Node.js, Express.js, REST APIs, MongoDB, PostgreSQL                                                                                                                  |
| <b>ML Frameworks &amp; Tools:</b> | TensorFlow, Keras, Git, GitHub, Visual Studio Code                                                                                                                   |
| <b>Soft Skills:</b>               | Problem Solving, Analytical Thinking, Team Collaboration, Communication, Agile Development                                                                           |